

MIDDLETOWN, OH, USA

WMO: 722249

Lat: 39.531N

Lon: 84.395W

Elev: 198

StdP: 98.97

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 63887

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-12.0	-20.7	0.6	-12.6	-17.6	0.8	-10.0	10.8	3.5	9.4	1.1	1.4	230	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	34.8	24.8	32.6	23.5	32.1	23.2	26.3	31.9	25.3	30.9	24.3	28.9	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.5	29.2	23.8	19.1	27.5	22.8	18.0	26.7	83.1	31.8	78.2	31.1	74.1	29.5	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	7.1	DB	-19.2	35.5	4.5	1.5	-22.4	36.6	-25.0	37.4	-27.5	38.2	-30.7	39.3	(4)
(5)			WB	-19.4	28.0	4.2	1.1	-22.4	28.8	-24.8	29.4	-27.2	30.0	-30.2	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-0.8	0.4	6.4	12.6	18.4	22.5	24.1	23.8	20.5	13.7	6.7	2.6	(6)
(7)		DBStd	10.16	6.70	6.26	5.81	5.07	4.30	3.19	2.94	2.99	4.12	5.15	4.92	5.46	(7)
(8)		HDD10.0	1168	338	274	143	31	1	0	0	0	0	21	122	238	(8)
(9)		HDD18.3	2731	592	502	372	181	53	5	1	1	24	162	349	488	(9)
(10)		CDD10.0	2132	4	5	31	108	263	377	436	427	316	134	23	8	(10)
(11)		CDD18.3	651	0	0	1	8	57	131	178	170	89	18	0	0	(11)
(12)		CDH23.3	6238	0	1	13	117	562	1212	1681	1595	868	187	2	0	(12)
(13)		CDH26.7	2361	0	0	1	16	157	459	682	639	356	53	0	0	(13)
(14)	Wind	WSAvg	2.6	3.3	3.2	3.1	3.2	2.5	2.3	1.9	1.7	1.8	2.4	2.5	2.9	(14)
(15)	Precipitation	PrecAvg	1034	81	54	84	100	121	111	102	78	73	77	72	82	(15)
(16)		PrecMax	1448	191	117	195	227	232	192	216	127	158	160	127	220	(16)
(17)		PrecMin	766	34	21	27	35	32	29	61	38	18	22	21	26	(17)
(18)		PrecStd	159	36	27	36	45	56	40	40	27	43	40	29	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	20.4	25.1	27.9	32.5	35.0	35.9	36.2	35.2	32.6	23.8	20.1	(19)
(20)			MCWB	15.9	14.6	16.7	17.5	22.9	25.3	26.2	26.7	22.5	22.6	16.1	16.6	(20)
(21)		2%	DB	15.3	16.0	22.3	26.7	30.4	32.7	33.6	34.0	32.7	27.4	20.7	16.0	(21)
(22)			MCWB	12.8	12.3	16.2	17.0	21.8	23.2	25.3	23.9	22.5	18.7	14.7	13.2	(22)
(23)		5%	DB	12.6	12.9	19.0	24.4	28.0	32.0	32.3	32.3	31.6	25.1	17.7	13.8	(23)
(24)			MCWB	10.7	9.7	12.8	16.0	20.3	22.6	24.1	23.7	22.3	18.0	13.7	11.8	(24)
(25)		10%	DB	9.4	9.8	16.0	22.4	26.8	29.3	31.1	31.0	27.9	22.8	15.4	11.8	(25)
(26)			MCWB	7.0	6.9	11.2	15.5	19.7	21.6	24.0	22.6	19.9	17.3	10.8	9.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	16.6	18.9	20.7	24.4	26.5	28.0	28.0	25.2	23.1	18.8	16.8	(27)
(28)			MCD B	18.7	20.0	23.8	25.3	30.5	32.4	33.2	33.8	30.4	29.8	21.5	18.9	(28)
(29)		2%	WB	13.0	13.4	16.4	19.0	22.9	25.1	26.6	26.1	24.1	21.5	16.3	14.0	(29)
(30)			MCD B	14.4	15.3	21.0	23.4	27.9	31.4	31.9	31.9	29.3	25.4	19.2	15.4	(30)
(31)		5%	WB	10.8	10.4	14.2	17.6	22.0	24.1	25.5	24.9	23.1	19.7	13.7	12.1	(31)
(32)			MCD B	12.6	12.9	17.3	22.2	26.6	28.9	30.9	30.1	28.1	22.9	16.9	13.3	(32)
(33)		10%	WB	7.2	7.0	12.3	16.2	21.0	23.0	24.4	23.8	22.2	18.2	12.0	9.7	(33)
(34)			MCD B	9.0	9.8	15.2	20.7	25.4	27.3	28.6	28.3	27.0	21.9	14.8	11.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	9.3	10.8	12.7	11.9	11.6	11.1	11.8	12.6	12.1	11.1	8.4	(35)
(36)			MCD BR	11.5	13.8	15.5	15.7	14.3	14.3	13.0	13.6	15.7	14.8	15.1	11.5	(36)
(37)			MCWBR	9.7	10.7	9.4	8.5	7.0	6.6	6.0	5.9	6.5	7.4	10.1	10.0	(37)
(38)		5% WB	MCD BR	10.7	12.7	12.4	12.8	12.9	12.3	11.6	11.9	12.7	11.8	12.7	10.5	(38)
(39)			MCWBR	9.6	10.8	9.0	8.5	6.8	6.5	5.9	5.7	7.3	9.8	9.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.328	0.352	0.398	0.442	0.453	0.456	0.451	0.400	0.358	0.328	0.305	(40)	
(41)		taud	2.455	2.359	2.366	2.252	2.170	2.189	2.187	2.183	2.302	2.419	2.462	2.469	(41)	
(42)		Ebn at Noon	866	889	902	877	842	830	824	817	840	844	830	837	(42)	
(43)		Edh at Noon	80	101	111	133	148	145	144	141	118	94	79	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.75	2.57	3.60	4.85	5.49	6.08	5.91	5.46	4.49	3.19	2.16	1.48	(44)	
(45)		RadStd	0.15	0.31	0.31	0.39	0.46	0.38	0.36	0.28	0.43	0.30	0.25	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+74		

Nomenclature: See separate page